

Department of Mathematics and Applied Mathematics

3MS Metric Spaces

Tutorial 3

2024

1. For each of the subsets A of the metric spaces listed below, answer the following, providing brief proofs. (i) Is A closed? (ii) What is $\overline{A}$?
 - (a) $A = \{x \in \mathbb{R} : |x| \geq 2\}$ in the reals with the usual metric.
 - (b) $A = \{x \in \mathbb{R} : 0 \leq x \leq 1 \text{ and } x \text{ is irrational}\}$ in the reals with the usual metric.
 - (c) $A = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ in $\mathbb{R}^2$ with the usual metric.
2.
 - (a) In $\mathbb{R}^2$ with the usual metric, is the open ball $B((0, 0); 1)$ a closed set? Justify your answer. (This is easy; it's just setting the scene for the next question.)
 - (b) In lectures, we proved that every closed ball in a metric space is a closed set. That is perfectly correct. We now adapt the proof slightly to try to show that every open ball is a closed set. This is false. Can you find the mistake in the "proof" below?

Incorrect statement In a metric space, every open ball is a closed set.

Incorrect argument Begin with a sequence (x_n) in the metric space (X, d) , such that $x_n \rightarrow x$ in (X, d) .

Consider an open ball $B(a; r)$, where $a \in X$ and $r > 0$; suppose that, for all n , $x_n \in B(a; r)$.

We show that $x \in B(a; r)$.

For all n , $d(a, x) \leq d(a, x_n) + d(x_n, x) < r + d(x_n, x)$.

Take limits as $n \rightarrow \infty$ on both sides, and use $\lim d(x_n, x) = 0$.

Then $d(a, x) < r + 0 = r$, as required.

- (c) Ultrametrics tend to provide examples where non-intuitive results hold. This is a case in point. Recall that an ultrametric space is a metric space (X, d) in which this strengthened form of the triangle inequality holds: for any $x, y, z \in X$,

$$d(x, z) \leq \max\{d(x, y), d(y, z)\}.$$

Prove that, in an ultrametric space, every open ball is a closed set.

Please make sure that you can see why your proof for the ultrametric case above would not work in every metric space.

3. Let d be the metric on $\mathbb{C}$ defined by

$$d(z, w) = \begin{cases} 0 & \text{if } z = w \\ |z| + |w| & \text{if } z \neq w. \end{cases}$$

In Tutorial 2 you were asked to characterise the sequences in $\mathbb{C}$ that converge with respect to this metric. You will need that answer now. They were (1) the eventually constant sequences, and (2) those (z_n) for which $|z_n| \rightarrow 0$ (using the usual metric on $\mathbb{C}$).

- (a) Is the set $\{z \in \mathbb{C} : |z| > 1\}$ closed with respect to this metric? And the set $\{z \in \mathbb{C} : 0 < |z| < 1\}$? Give reasons for your answers.
 (b) Write down the closures of the sets in (a).

For hints on how to work with suprema and infima, including some common misconceptions, please see the note entitled “Real analysis- suprema and infima” under Vula notes resources.

4. Recall from Tutorial 2 that we define, for non-empty subsets A, B of a metric space (X, d) ,

$$\text{dist}(A, B) = \inf\{d(a, b) \mid a \in A, b \in B\}.$$

Read the theorem and proof below and then answer the questions that follow.

Theorem For any non-empty subsets A and B of a metric space (X, d) ,

$$\text{dist}(A, B) = \text{dist}(\overline{A}, B).$$

Proof: Since $A \subseteq \overline{A}$, we have $\text{dist}(A, B) \geq \text{dist}(\overline{A}, B)$. (1)

Let ϵ be an arbitrary positive real number. (2)

There exists $c \in \overline{A}$ and $b \in B$ such that $d(c, b) \leq \text{dist}(\overline{A}, B) + \frac{\epsilon}{2}$. (3)

There exists $a \in A$ such that $d(a, c) \leq \frac{\epsilon}{2}$. (4)

Then $\text{dist}(A, B) \leq d(a, b) \leq d(a, c) + d(c, b) \leq \text{dist}(\overline{A}, B) + \epsilon$. (5)

Since ϵ was arbitrary, we obtain $\text{dist}(A, B) \leq \text{dist}(\overline{A}, B)$. (6)

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- (a) Show that, if C and D are both non-empty subsets of the reals, then $C \subseteq D$ implies that $\inf C \geq \inf D$. (Use the definition of “infimum” and nothing else!) Use this to justify the claim in line (1).
 (b) Explain why b and c can be obtained as claimed in line (3).
 (c) Explain where a comes from, in line (4).
 (d) Justify each of the inequalities in line (5).
 (e) In order to see how line (6) follows from line (5), prove the following: If x, y are real numbers for which $x \leq y + \epsilon$ for all real $\epsilon > 0$, then $x \leq y$.
 (f) Deduce the corollary: For any non-empty subsets A and B of a metric space (X, d) , $\text{dist}(A, B) = \text{dist}(\overline{A}, B)$.